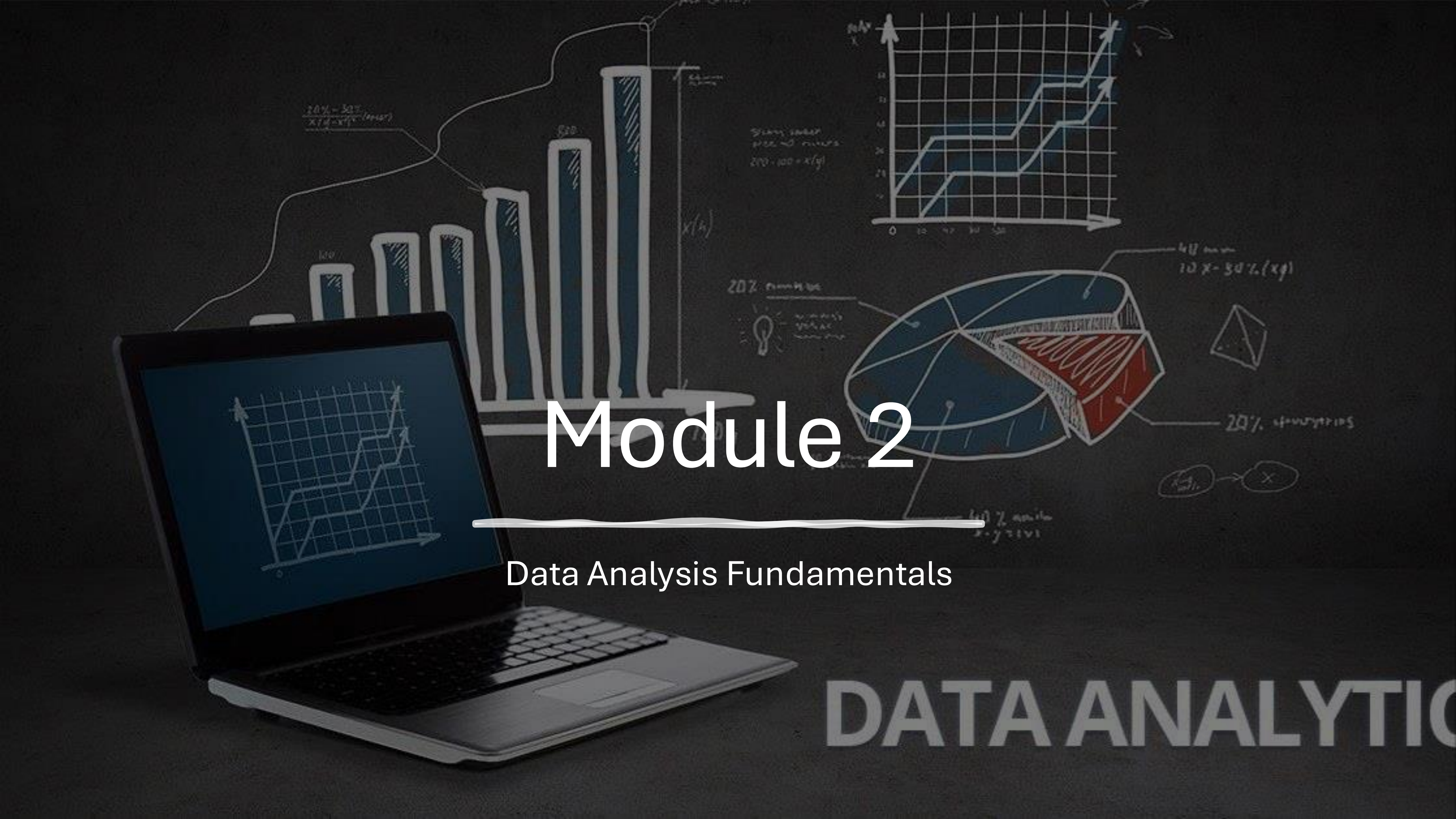
K-Youth Training Program

Data Analytics for business growth and sustainability in the workforce



Module 2

Data Analysis Fundamentals

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Module Contents

Module 2: Data Analysis Fundamentals

- 1. Databases – types and purpose
- 2. Tools used to ingest data
- 3. Storing and retrieving data
- 4. Data Quality and Cleaning Techniques
- 5. Data Transformation
- 6. Data Integration and ETL Processes
- 7. Python Basics for Non-IT Students
- a. Introduction to Python and Its Uses in Data Analytics
- b. Setting Up Python Environment
- c. Basic Python Syntax and Operations
- d. Data Structures: Lists, Tuples, Dictionaries
- e. Importing and Using Libraries (NumPy, Pandas)
- f. Basic Data Manipulation with Pandas

Data Ingestion Pipeline

Data Sources

- Batch
- Streaming
- Types
  - API
  - JDBC
  - ODBC

Data Ingestion

- Batch
  - Scheduled
  - Bulk data
- Stream
  - Kafka
  - MQTT

Storage / processing

- Big data - HDFS
- RDBMS
- Document DB
- Graph DB
- Data Lake
- Data warehouse

Transformation / modeling

- Cleaning
- Filtering
- Aggregation
- Transformation
- Enrichment
- Workflows

Consumption

- BI Analytics
- Machine learning
- Data products

Data Observability and Monitoring

Security and Governance

Data sources

Data Sources

- Batch
- Streaming
- Types
  - API
  - JDBC
  - ODBC

Data can come in many forms generally categorized as Structured, Semi-Structured and Unstructured

- Sample data types are as below:
  - Excel
  - Text – TXT, CSV, TSV
  - PDF
  - Image / Video / Audio
  - JSON
  - Streaming data
  - Databases

Data Ingestion

Data Ingestion

- Batch
  - Scheduled
  - Bulk data
- Stream
  - Kafka
  - MQTT

Process of extracting the data from the sources and feeding into the pipeline

- Generally, data is consumed in the following methods
  - Batch
  - Streaming
- Batch ingestion is a scheduled or adhoc process for bulk data transfer
- Streaming ingestion is usually for continuous data extraction with support for multiple types of formats
- Error handling, retries are methods to ensure resilience in data extraction process

Storage and processing

Storage / processing

- Big data - HDFS
- RDBMS
- Document DB
- Graph DB
- Data Lake
- Data warehouse

Storage area of the data for further processing or querying

- Generally data ingested can be stored based on the usage of the data
  - **Data Lakes** – a scalable storage for raw, unstructured and semi-structured data. Data stored as raw are usually used for further processing step like cleaning,filtering etc.
  - **Data Warehouse** – for structured data than can be queried with ease. Data here is generally already structured to answer business questions.
- Common Tools include
  - Amazon S3, Azure Data Lake, Snowflake, BigQuery

Transformation / Modeling

Transformation / modeling

- Cleaning
- Filtering
- Aggregation
- Transformation
- Enrichment
- Workflows

- Generally the preparation of Data for analysis occurs at this step
- Common activities include
  - Data cleansing – removing of duplicates, empty rows, outliers, standardizing data types etc.
  - Data Enrichment – adding value to the data by integrating with other data sources
  - Transformation – converts data into the desired format or schema
  - Stream processing – apply real time transformation to raw data
- Common tools include
  - Python
  - Commercial tools
  - Apache Flink
  - Spark Streaming

Data Consumption

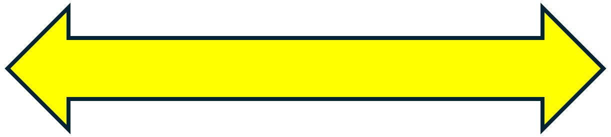
Consumption

- BI Analytics
- Machine learning
- Data products

The endpoint where data is delivered for consumption to end users or other systems

- Support for data warehouses, lakes, and real-time dashboards.
- Integration with business intelligence tools (e.g., Tableau, Power BI).
- Support for machine learning platforms or custom applications.
- API-driven or batch-based data export capabilities.

Data Observability and Monitoring



Data Observability and Monitoring

Ensures the health, performance, and accuracy of the pipeline.

- Real-time metrics and alerts for latency, throughput, and failures.
- Data quality checks (e.g., schema validation, anomaly detection).
- Centralized logging systems (e.g., ELK Stack, Datadog).
- End-to-end lineage tracking for compliance and debugging.

Security and Governance

Security and Governance

Ensures that the pipeline complies with organizational policies and regulations.

- **Authentication and Authorization**: Role-based access control (RBAC).
- **Data Encryption**: In transit and at rest.
- **Data Masking**: Obscuring sensitive information for compliance.
- **Audit Trails**: Tracking data access and modifications.
- **Ethical Governance**: Ensures data is used for the designated purpose – HIPAA etc.